



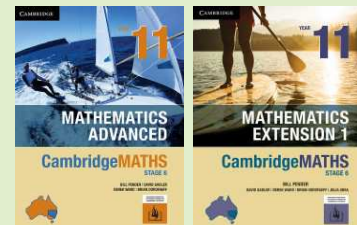
**NORMANHURST BOYS HIGH SCHOOL**

# **MATHEMATICS ADVANCED (INCORPORATING EXTENSION 1) YEAR 11 COURSE**

 **Topic summary and exercises:**

**(A) (x1) Indices and Logarithms**

With references to



Name: .....

Initial version by H. Lam, September 2014 (Graphs of exponential/logarithmic functions). Last updated September 20, 2023. Various corrections by students & members of the Department of Mathematics at North Sydney Boys and Normanhurst Boys High Schools.

**Acknowledgements** Pictograms in this document are a derivative of the work originally by Freepik at <http://www.flaticon.com>, used under  CC BY 2.0.

## Symbols used

-  Beware! Heed warning.
-  Mathematics Advanced content.
-  Mathematics Extension 1 content.
-  Literacy: note new word/phrase.
- $\mathbb{N}$  the set of natural numbers
- $\mathbb{Z}$  the set of integers
- $\mathbb{Q}$  the set of rational numbers
- $\mathbb{R}$  the set of real numbers
- $\forall$  for all

## Syllabus outcomes addressed

**MA11-6** manipulates and solves expressions using the logarithmic and index laws, and uses logarithms and exponential functions to solve practical problems

## Syllabus subtopics

**MA-E1** (1.1, 1.2, 1.4) Logarithms and Exponentials

### Gentle reminder

- For a thorough understanding of the topic, *every* question in this handout is to be completed!
- Additional questions from *CambridgeMATHS Mathematics Extension 1* (Pender, Sadler, Ward, Dorofaeff, & Shea, 2019b) or *CambridgeMATHS Mathematics Advanced* (Pender, Sadler, Ward, Dorofaeff, & Shea, 2019a) will be completed at the discretion of your teacher.
- Remember to copy the question into your exercise book!

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# Section 1

## Ⓡ Review of index laws



### Learning Goal(s)



#### Knowledge

What are the index laws



#### Skills

Applying the index laws to simplify expressions



#### Understanding

How to derive the index laws



By the end of this section am I able to:

6.1 Use the index laws

### 1.1 Operations resulting in addition/subtraction/multiplication of indices

Only the most challenging questions are provided here. The rest are omitted for brevity and should be assumed knowledge.



### Laws/Results

For  $a > 1$ :

$$\text{(Product)} \quad a^x \times a^y = \dots\dots\dots (1.1)$$

$$\text{(Quotient)} \quad a^x \div a^y = \dots\dots\dots (1.2)$$

$$\text{(Power)} \quad (a^x)^y = \dots\dots\dots (1.3)$$

$$\text{(Power 0)} \quad a^0 = \dots\dots\dots (1.4)$$

$$\text{(Negative index)} \quad a^{-x} = \dots\dots\dots (1.5)$$

$$\text{(Fractional index)} \quad a^{\frac{1}{x}} = \dots\dots\dots (1.6)$$



### Laws/Results

(A slightly more neglected rule)

$$(ab)^x = \dots\dots\dots (1.7)$$

 **Important note**

Use these laws in either “direction”, i.e. from L to R *or* R to L

 **Example 1**

[Ex 8A Q15(e)] Write as a single fraction, without negative indices and simplify:

$$x^{-2}y^{-2}(x^2y^{-1} - y^2x^{-1})$$

 **Important note**

An everyday, plain English definition of *simplify*: write the expression with .....  
.....

 **Example 2**

Fully simplify:  $\frac{2^{n+1} - 2^n}{2^{2n+1} - 2^{2n}}$

**Answer:**  $2^{-n}$

 **Example 3**

[Ex 8A Q16(f)] Fully simplify:

$$\frac{24^{x+1} \times 8^{-1}}{6^{2x}}$$

**Answer:**  $2^x 3^{1-x}$

**Example 4**

[Ex 8A Q18(c)] Fully simplify:  $\frac{12^n - 18^n}{3^n - 2^n}$ .

**Answer:**  $-6^n$

## 1.2 Fractional indices

**Example 5**

Fully simplify:  $\sqrt[3]{\frac{x^{5k+1}y^{4k-5}}{x^{2(k-1)}y^{-2k+1}}}$

**Answer:**  $x^{k+1}y^{2(k-1)}$

**Further exercises**

**Ex 8A**

- Q5-19

**Ex 8B**

- Q5-19, 21

# Section 2

## Ⓡ Logarithms



### Learning Goal(s)

#### 📖 Knowledge

What are the logarithmic laws

#### ⚙️ Skills

Applying the logarithm laws to simplify expressions

#### 💡 Understanding

How to derive the logarithmic laws

#### ✅ By the end of this section am I able to:

- 6.2 Define logarithms as indices:  $y = a^x$  is equivalent to  $x = \log_a y$ , and explain why this definition only makes sense when  $a > 0$  and  $a \neq 1$
- 6.3 Understand how to use a calculator to find logarithms base 10.
- 6.4 Recognise and use the inverse relationship between logarithms and exponentials.
- 6.5 Derive the logarithmic laws from the index laws and use the algebraic properties of logarithms to simplify and evaluate logarithmic expressions. exponentials.

### 2.1 Basic rules

#### 📖 Definition 1

If  $a^x = y$ , then  $x = \log_a y$ .

### 2.2 Laws of logarithms

#### 🔴 Laws/Results

$$\text{(Product)} \quad \log x + \log y = \log ( \quad ) \quad (2.1)$$

$$\text{(Quotient)} \quad \log x - \log y = \log \left( \quad \right) \quad (2.2)$$

$$\text{(Power)} \quad \log (x^a) = \dots\dots\dots (2.3)$$

$$\text{(Power 1)} \quad \log_a a = \dots (2.4)$$

$$\text{(Power 0)} \quad \log_a 1 = \dots (2.5)$$

**Important note**

Use these laws in either “direction”, i.e. from L to R *or* R to L

**Example 6**

Given  $\log_a 2 = m$  and  $\log_a 3 = n$ , write the following in terms of  $m$  and  $n$ :

- |                 |                  |                   |
|-----------------|------------------|-------------------|
| (a) $\log_a 4$  | (c) $\log_a 1.5$ | (e) $\log_a 18$   |
| (b) $\log_a 27$ | (d) $\log_a 12$  | (f) $\log_a 13.5$ |

**Example 7**

[1995 2U HSC Q7] Given  $\log_a b = 2.75$  and  $\log_a c = 0.25$ , find the value of

- |                                     |          |
|-------------------------------------|----------|
| i $\log_a \left(\frac{b}{c}\right)$ | <b>1</b> |
| ii $\log_a (bc)^2$                  | <b>2</b> |

**Example 8**

Write each of the following as  $\log_2 3$ :

- |                 |                        |                          |
|-----------------|------------------------|--------------------------|
| (a) $\log_2 81$ | (b) $\log_2 2\sqrt{3}$ | (c) $\log_2 \frac{8}{9}$ |
|-----------------|------------------------|--------------------------|

## 2.3 Exponential analog & mutual inverse

### Laws/Results

(Mutual inverse)

$$a^{\log_a x} = \dots \quad \log_a a^x = \dots \quad (2.6)$$

### Example 9

Express 5 as a power of 2.

### Example 10

[Ex 8D Q15] Fully simplify:  $2^{\log_2 3 + \log_2 5}$

### Further exercises

**Ex 8C** (Pender et al., 2019b)

- Q5-11

**Ex 8D** (Pender et al., 2019b)

- Q5-16

**Ex 12A** (Pender, Sadler, Shea, & Ward, 1999)

- Q3-16, **(E)** Q17-18

## Section 3

# Equations involving indices and logarithms



### Learning Goal(s)

#### Knowledge

What is the change of base law

#### Skills

Applying the logarithm laws to solve equations

#### Understanding

How to make a pronumeral the subject in equations involving exponentials and logarithms

#### By the end of this section am I able to:

- 6.7 Consider different number bases and prove and use the change of base law  $\log_a x = \frac{\log_b x}{\log_b a}$
- 6.8 Can solve a range of equations involving indices and logarithms

### 3.1 Change of base



#### Steps

To change from base  $a$  for  $\log_a x$  to base  $b$ :

1. Let  $y = \log_a x$ .

2. Rewrite as exponential:

.....

3. Take logarithm to base  $b$ :

.....

4. Use log-power law (2.3):

.....

5. Divide and rewrite:

.....

#### 3.1.1 Examples



#### Example 11

Solve  $2^x = 7$  correct to 4 significant figures.

Answer: 2.807

**Example 12****[2010 NSB Ext 1 Prelim Yearly]**

- |    |                                                                                                |          |
|----|------------------------------------------------------------------------------------------------|----------|
| i  | Rewrite $2^x = 5^y$ with $\frac{x}{y}$ as the subject by taking logarithms to a suitable base. | <b>1</b> |
| ii | Hence or otherwise, find the exact value of $8^{\frac{x}{y}}$ if $2^x = 5^y$ .                 | <b>2</b> |

**Answer:** (i)  $\log_2 5$  (ii) 125

## 3.2 Harder equations

### 3.2.1 Exponential



#### Example 13

Solve for  $x$ :

$$3^{2x} - 12 \times 3^x + 27 = 0$$

**Answer:** 1, 2



#### Example 14

Solve for  $x$ :

$$4^x - 12 \times 2^x + 32 = 0$$

**Answer:** 2, 3

### 3.2.2 Logarithmic



#### Example 15

Solve:  $2\log_3 x = \log_3(6 - x)$ .

**Answer:** 2



#### Example 16

Solve:  $\log 16 - \log x = \log(8 - x)$



#### Example 17

[2019 NBHS Ext 1 Task 2] Find the value of  $x$  if

$$\log_a 2 + 2\log_a x - \log_a 6 = \log_a 3$$

**Answer:** 3

**Example 18**

Solve the following pair of simultaneous equations:

$$\begin{cases} 5^{x+y} = \frac{1}{5} \\ 5^{3x+2y} = 1 \end{cases}$$

**Answer:**  $x = 2, y = -3$

**Example 19**

(Pender et al., 1999, Ex 12A)

- (a) Use the change of base rule to show that

$$\log_{a^x} b = \frac{\log_a b}{x}$$

- (b) Hence evaluate  $\log_{\sqrt{27}} 81$  without a calculator.

- (c) Solve for  $x$ :

$$\log_{\sqrt{a}}(x+2) - \log_{\sqrt{a}} 2 = \log_a x + \log_a 2$$

**Answer:** (b)  $\frac{8}{3}$  (c)  $x = 2$

**Example 20**

[Ex 12A Q18] (Pender et al., 1999)  Solve for  $x$ :  $\log_{2x} 216 = x$

**Answer:** 3

**Important note**

Hint: Rewrite in index form, then consider the prime factorisation of 216.

**Further exercises**

**Ex 8B**

- Q20

**Ex 8E**

- Q2, 4-15,  Q16

## Section 4

# Exponential and logarithmic graphs

### 4.1 Exponential



#### Learning Goal(s)

##### Knowledge

What are the important features of exponential and logarithmic graphs

##### Skills

Sketch the graphs of exponentials and logarithms

##### Understanding

The techniques used to determine the important features on exponential and logarithmic graphs

##### ✓ By the end of this section am I able to:

6.10 Recognise and sketch the graphs of  $y = ka^x$ ,  $y = ka^{-x}$  where  $k$  is a constant, and  $x = \log_a y$

6.11 Solve a range of problems related to exponential and logarithmic functions



#### Theorem 1

(M) Basic exponential curves:

- Equation: .....
- Condition on  $a$ : .....
- Domain: .....
- Range: .....
- When  $x = 0$ , .....
- When  $x = 1$ , .....
- As
  - $\lim_{x \rightarrow \infty} \dots = \dots$
  - $\lim_{x \rightarrow -\infty} \dots = \dots$
- Inverse: .....

Sketch:

- Always show two points on the curve!

## 4.1.1 Transformations



## Example 21

Sketch:

1.  $y = 3^x$

3.  $y = 10^x$

5.  $y = 2^{x-1}$

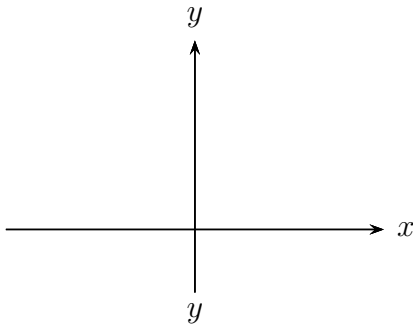
7.  $y = 1 - 2^{-x}$

2.  $y = 3^{-x}$

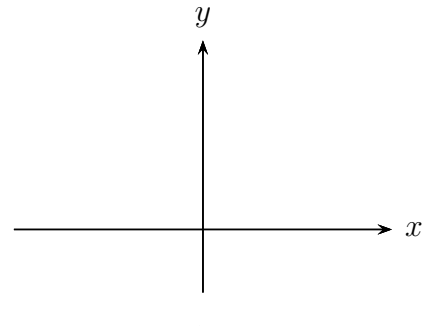
4.  $y = \left(\frac{1}{6}\right)^x$

6.  $y = 2^x - 1$

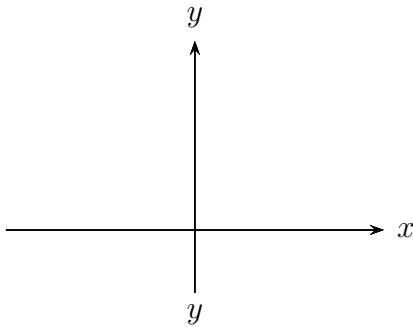
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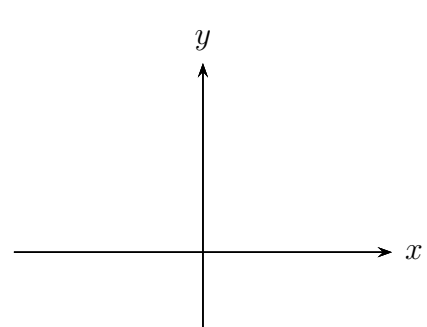
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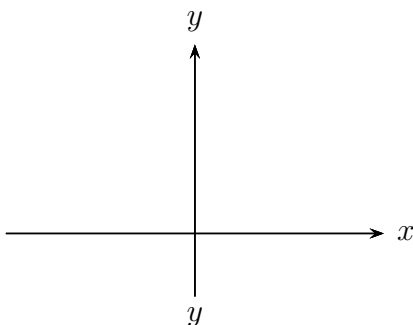
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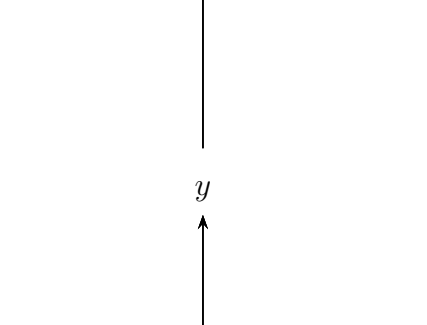
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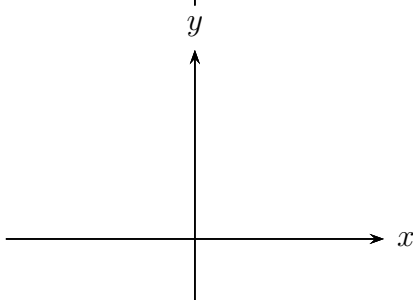
3.



7.



4.



### 4.1.2 (x1) Harder graphs



#### Example 22

Sketch  $y = 2^{|x|}$

**Example 23**

Sketch  $y = 2^x + 2^{-x}$

## 4.2 Logarithmic

### Theorem 2

Basic logarithmic curves:

- Equation: .....
- Condition on  $a$ : .....
- Domain: .....
  - $x \rightarrow \infty, y \rightarrow \dots$
  - $x \rightarrow 0^+, y \rightarrow \dots$
- Range: .....
- When  $x = 1$ , .....
- When  $x = a$ , .....
- As .....
- Inverse: .....

**Sketch:**

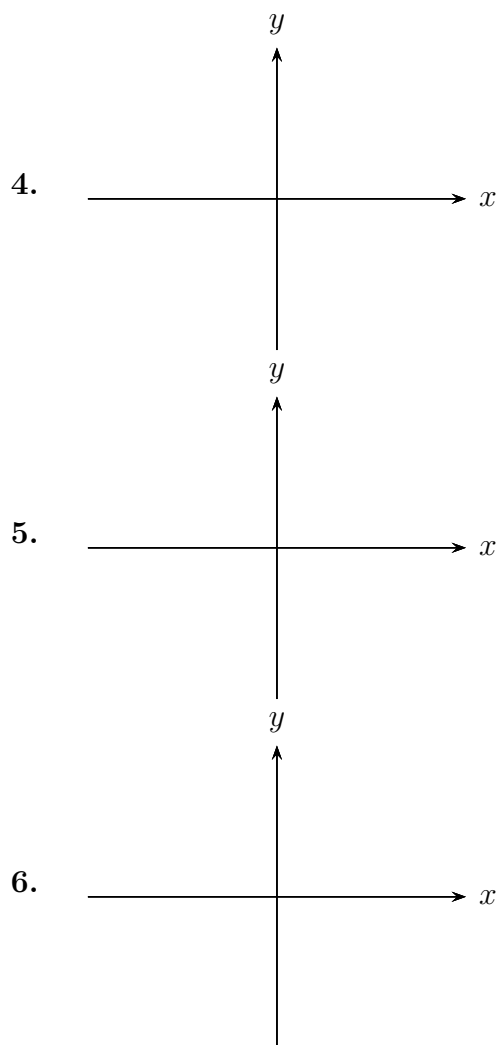
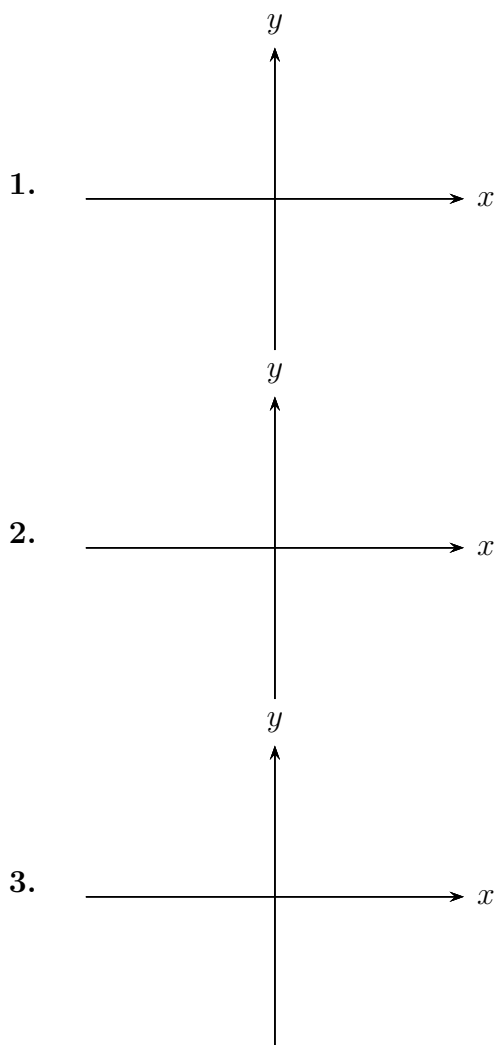
- Always show two points on the curve!

### 4.2.1 Transformations



#### Example 24

- |                                |                             |
|--------------------------------|-----------------------------|
| 1. $f(x) = \log_{10}(2x).$     | 4. $f(x) = \log_7(2x + 3).$ |
| 2. $f(x) = \log_5 x^2, x > 0.$ | 5. $f(x) = \log_{10}  x .$  |
| 3. $f(x) = \log_6(3x - 6).$    | 6. $f(x) = \log_7(-x).$     |



#### Further exercises

Ex 8F

- Q2-8

### 4.2.2 (x1) Harder graphs



#### Example 25

Sketch  $y = \log_5 \sqrt{x}$



#### Example 26

Sketch  $y = \log_3 |x|$



#### Example 27

Sketch  $|y| = \log_{10}(-x)$

**Example 28**

Sketch  $y^2 = \log_{10}(-x)$

**Further exercises**

**Ex 8F**

- Q9-11

# Section 5

## Applications



### Learning Goal(s)



#### Knowledge

Solving real-life problems using exponentials and logarithms



#### Skills

Constructing equations involving exponentials and logarithms to solve from real life contexts



#### Understanding

The real life applications of exponentials and logarithms

#### ✓ By the end of this section am I able to:

- 6.6 Interpret and use logarithmic scales, for example decibels in acoustics, different seismic scales for earthquake magnitude, octaves in music or pH in chemistry
- 6.9 Solve algebraic, graphical and numerical problems involving logarithms in a variety of practical and abstract contexts, including applications from financial, scientific, medical and industrial contexts



### Important note

Most questions in this section make use of the exponent  $e$  instead of 2, 3 or 10. Simply treat  $e$  as another one of these numerals for the time being - there will be a more in depth study of  $e$  later on with calculus.



### Example 29

[2019 CSSA 2U Trial Q16] (2 marks) Find the domain of the function

$$g(x) = \ln(x^2 - x)$$

**Example 30**

[2000 2U HSC Q9] with slight modifications

- |     |                                                                                |          |
|-----|--------------------------------------------------------------------------------|----------|
| i.  | Sketch $y = \log_e x$ (Note: $e \approx 2.71828 \dots$ )                       | <b>2</b> |
| ii. | On the same sketch, find, graphically, the number of solutions of the equation | <b>1</b> |

$$\log_e x - x = -2$$

**Example 31**

[1990 2U HSC Q7] A quantity  $Q$  of radium at time  $t$  in years is given by

$$Q = Q_0 e^{-kt}$$

where  $k$  is a constant and  $Q_0$  is the initial amount of radium at time  $t = 0$ .

- i. Given that  $Q = \frac{1}{2}Q_0$ , when  $t = 1\,690$  years, calculate the value of  $k$  **1**
- ii. After how many years does only 20% of the initial amount of radium remain? **2**

**Example 32**

[2020 CSSA Adv Trial Q18] A cup of coffee is cooling according to the following exponential formula

$$C = 21 + (74 \times 3^{-0.2t})$$

where  $C$  is the temperature in degrees Celsius and  $t$  is the time in minutes since the coffee was poured.

- (a) Calculate the initial temperature of the coffee. **1**
- (b) Calculate the temperature of the coffee after 10 minutes, correct to the nearest degree. **1**
- (c) After how many minutes, to the nearest minute, will the coffee first reach  $50^\circ\text{C}$ ? **2**

**Answer:** (a)  $95^\circ\text{C}$  (b)  $29.2^\circ\text{C}$  (c)  $\approx 4$  min 16 seconds

**Example 33**

[1996 2U HSC Q3] A layer of plastic cuts out 15% of the light and lets through the remaining 85%.

- i. Show that two layers of the plastic let through 72.25% of the light 1
- ii. How many layers of the plastic are required to cut out at least 90% of the light? 2

**Example 34**

[1985 2U HSC Q10] A population  $N(t)$  varies with time according to the law

$$N(t) = Ce^{kt}$$

where  $C, k$  are positive constants and  $t \geq 0$

- i. Show that, if  $a, b$  are two positive numbers such that  $a + b = 1$ , then 1

$$N(at + bu) = \left(N(t)\right)^a \left(N(u)\right)^b$$

for any  $t \geq 0, u \geq 0$ .

- ii. Hence or otherwise, find  $N(13)$ , given that  $N(3) = 10$  and  $N(18) = 100$  2

**Example 35**

[2011 2U HSC Q10] The intensity  $I$ , measured in watt/m<sup>2</sup>, of a sound is given by

$$I = 10^{-12} \times e^{0.1L}$$

where  $L$  is the loudness of the sound in decibels.

- |    |                                                                                                                                    |          |
|----|------------------------------------------------------------------------------------------------------------------------------------|----------|
| i  | If the loudness of a sound at a concert is 110 decibels, find the intensity of the sound. Give your answer in scientific notation. | <b>1</b> |
| ii | Ear damage occurs if the intensity of a sound is greater than $8.1 \times 10^{-9}$ watt/m <sup>2</sup> .                           | <b>2</b> |

What is the maximum loudness of a sound so that no ear damage occurs?

- |     |                                                                                       |          |
|-----|---------------------------------------------------------------------------------------|----------|
| iii | By how much will the loudness of a sound have increased if its intensity has doubled? | <b>2</b> |
|-----|---------------------------------------------------------------------------------------|----------|

**Example 36**

[1990 2U HSC Q8] It is assumed that the number  $N(t)$  of termites in a certain mound at time  $t \geq 0$  is given by

$$N(t) = \frac{A}{2 + e^{-t}}$$

where  $e \approx 2.7818 \dots$ ,  $A$  is a constant and  $t$  is measured in months.

- i. At time  $t = 0$ ,  $N(t)$  is estimated at  $3 \times 10^5$  termites. What is the value of  $A$ ? 1
- ii. What is the value of  $N(t)$  after one month? 1
- iii. How many termites would you expect to find in the mound when  $t$  is very large? 1
- iv. ((x1) - added and modified) By considering the graph of  $y = 2 + e^{-t}$ , sketch the graph of  $N(t)$ .

**Further exercises**

**Ex 8G**

- All questions

### 5.1 Additional questions

---

1. [1992 2U HSC Q4] Ten kilograms of sugar is placed in a container of water and begins to dissolve. After  $t$  hours the amount  $A$  kg of undissolved sugar is given by

$$A = 10e^{-kt}$$

(where  $e \approx 2.7818 \dots$ )

- i. Calculate  $k$ , given that  $A = 3.2$  when  $t = 4$ . **1**
  - ii. After how many hours does 1 kg of sugar remain undissolved? **1**
2. [2010 2U HSC Q4] (2 marks) Let  $f(x) = 1 + e^x$

Show that  $f(x) \times f(-x) = f(x) + f(-x)$ .

### Answers

1. i.  $k = \frac{1}{4} \ln \frac{10}{3.2} \approx 0.28$  ii.  $t \approx 8.08$  hrs

# References

- Pender, W., Sadler, D., Shea, J., & Ward, D. (1999). *Cambridge Mathematics 3 Unit Year 11* (1st ed.). Cambridge University Press.
- Pender, W., Sadler, D., Ward, D., Dorofaeff, B., & Shea, J. (2019a). *CambridgeMATHS Stage 6 Mathematics Advanced Year 11* (1st ed.). Cambridge Education.
- Pender, W., Sadler, D., Ward, D., Dorofaeff, B., & Shea, J. (2019b). *CambridgeMATHS Stage 6 Mathematics Extension 1 Year 11* (1st ed.). Cambridge Education.